MATH3303 – Lecture 14

Classification of Finite Abelian Groups V: The Payoff

Review Notes

Reference: Lee, Abstract Algebra*, §5.2–5.3*

Overview

- **Proof of the Fundamental Theorem of Finite Abelian Groups** (Lee, Thm. 5.3): uses Lemma 6 + Lemma 7 + induction.
- **Elementary divisors** (Lee, Def. 5.4): the canonical invariants of a finite abelian group.
- **Examples**: abelian groups of order 16 and order 20.
- **Cyclic product criterion** (Lee, Thm. 5.4): $H_1 \times \cdots \times H_k$ is cyclic iff all $|H_i|$ are pairwise coprime.
- **Classification Theorem** (Lee, Thm. 5.6): two finite abelian groups are isomorphic iff they have the same elementary divisors.
- **Extended example**: $G = \mathbb{Z}_{200} \times \mathbb{Z}_8 \times \mathbb{Z}_6$, $H = \mathbb{Z}_{120} \times \mathbb{Z}_{10} \times \mathbb{Z}_4 \times \mathbb{Z}_2$, $K = \mathbb{Z}_{25} \times \mathbb{Z}_{24} \times \mathbb{Z}_8 \times \mathbb{Z}_2$.

1 Proof of the Fundamental Theorem

Theorem 1.1: Fundamental Theorem of Finite Abelian Groups (from lecture; cf. Lee, Thm. 5.3)

Let G be a finite abelian group. Then G is the direct product of subgroups

$$G \;\cong\; H_1 \times H_2 \times \cdots \times H_r,$$

where each H_i is cyclic of order $p_i^{n_i}$, with p_i prime (not necessarily distinct) and $n_i \geq 1$.

Proof. If G is the trivial group, there is nothing to do.

Otherwise, by Lemma 6 (Lecture 12, Lee Lemma 5.5), G is the direct product of its p-subgroups. So WLOG assume G is a finite abelian p-group. We proceed by strong induction on $|G|$.

Base case $|G| = 1$: trivial.

Inductive step: Suppose the theorem holds for all abelian p-groups of smaller order. Let $g \in G$ be an element of largest order. By Lemma 7 (Lecture 13, Lee Lemma 5.6):

$$G = \langle g \rangle \times H \quad \text{for some subgroup } H \leq G.$$

Then $|H| = |G|/|\langle g \rangle| < |G|$, so H has smaller order. By the inductive hypothesis, H is a direct product of cyclic groups of prime power order. Since $\langle g \rangle$ is also a cyclic group of prime power order ($|g| = p^n$), G is as well. □

Remark 1.1: Corollary (Lee, Cor. 5.1)

Every nontrivial finite abelian group G is isomorphic to

$$\mathbb{Z}_{p_1^{n_1}} \times \mathbb{Z}_{p_2^{n_2}} \times \cdots \times \mathbb{Z}_{p_r^{n_r}},$$

where the p_i are primes (not necessarily distinct) and $n_i \geq 1$.

The numbers $p_1^{n_1}, p_2^{n_2}, \ldots, p_r^{n_r}$ are called the *elementary divisors* of G (see Definition 2.1 below). Order in this list is irrelevant, but each number must appear as many times as it occurs.

2 Elementary Divisors and Examples

Definition 2.1: Elementary divisors (from lecture; cf. Lee, Def. 5.4)

Let G be a nontrivial finite abelian group, written as

$$G \;\cong\; \mathbb{Z}_{p_1^{n_1}} \times \mathbb{Z}_{p_2^{n_2}} \times \cdots \times \mathbb{Z}_{p_r^{n_r}},$$

where each p_i is prime and $n_i \geq 1$. The *elementary divisors* of G are the numbers

$$p_1^{n_1},\; p_2^{n_2},\; \ldots,\; p_r^{n_r}.$$

Order is irrelevant, but each number is listed as many times as it appears.

Remark 2.1: Uniqueness (from lecture)

The elementary divisors uniquely determine a finite abelian group up to isomorphism (Classification Theorem, Section 3). So listing the elementary divisors is equivalent to classifying the group.

Example 2.1: Abelian groups of order 16 (from lecture; cf. Lee, Example 5.7)

$16 = 2^4$. The abelian groups of order 16 correspond to partitions of 4:

Group	Elementary divisors
$\mathbb{Z}_{16}$	16
$\mathbb{Z}_8 \times \mathbb{Z}_2$	$8, 2$
$\mathbb{Z}_4 \times \mathbb{Z}_4$	$4, 4$
$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	$4, 2, 2$
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	$2, 2, 2, 2$

These are all non-isomorphic (distinct elementary divisors) and every abelian group of order 16 is isomorphic to exactly one of them.

Example 2.2: Abelian groups of order 20 (from lecture)

$20 = 2^2 \times 5$. The abelian groups of order 20:

Group	Elementary divisors
$\mathbb{Z}_4 \times \mathbb{Z}_5$	$4, 5$
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5$	$2, 2, 5$

Why no $\mathbb{Z}_{20}$? Because $\mathbb{Z}_{20} \cong \mathbb{Z}_4 \times \mathbb{Z}_5$ (since $\gcd(4, 5) = 1$). So $\mathbb{Z}_{20}$ is already in the list.

3 The Cyclic Product Criterion

Theorem 3.1: Cyclic product criterion (from lecture; cf. Lee, Thm. 5.4)

Let $G = H_1 \times H_2 \times \cdots \times H_k$, where each H_i is cyclic of order n_i. Then G is cyclic if and only if $\gcd(n_i, n_j) = 1$ whenever $i \neq j$.

Proof. Write $H_i = \langle h_i \rangle$.

($\Leftarrow$) Suppose all n_i are pairwise coprime.

Claim: $|(h_1, h_2, \ldots, h_k)| = n_1 n_2 \cdots n_k = |G|$, so G is cyclic.

Suppose $(h_1, h_2, \ldots, h_k)^m = (1_G, 1_G, \ldots, 1_G)$. Then $h_i^m = 1_G$ for each i, so $n_i \mid m$ for all i. Since the n_i are pairwise coprime, $n_1 n_2 \cdots n_k \mid m$ (by Lee, Cor. 2.3). Since $|G| = n_1 n_2 \cdots n_k$, the largest possible element order is $n_1 n_2 \cdots n_k$, and we have found an element of this order. So G is cyclic.

($\Rightarrow$) Suppose some prime p divides both n_1 and n_2.

Claim: each $n_i \mid \dfrac{n_1 n_2 \cdots n_k}{p}$, so every element has order dividing $\dfrac{n_1 \cdots n_k}{p}$, and G is not cyclic.

For $i = 1$: $n_1 \mid \dfrac{n_1}{p} \cdot n_2 \cdots n_k = \dfrac{n_1 n_2 \cdots n_k}{p}$ (since $p \mid n_2$, we have $\frac{n_1 n_2 \cdots n_k}{p} = n_1 \cdot \frac{n_2}{p} \cdot n_3 \cdots n_k$, and n_1 divides this).

For $i \geq 2$: $n_i \mid \dfrac{n_1}{p} \cdot n_2 \cdots n_k = \dfrac{n_1 n_2 \cdots n_k}{p}$ (since $p \mid n_1$, we have $\frac{n_1}{p} \in \mathbb{Z}$, so n_i divides the product).

Therefore for any $(h_1^{r_1}, \ldots, h_k^{r_k}) \in G$:

$$\left(h_1^{r_1}, \ldots, h_k^{r_k}\right)^{\frac{n_1 \cdots n_k}{p}} = \left((h_1^{r_1})^{\frac{n_1 \cdots n_k}{p}}, \ldots\right) = (1_G, \ldots, 1_G),$$

since $n_i \mid \frac{n_1 \cdots n_k}{p}$ implies $h_i^{\frac{n_1 \cdots n_k}{p} \cdot r_i} = 1_G$.

So every element of G has order dividing $\frac{n_1 \cdots n_k}{p} < |G|$, and G has no element of order $|G|$. Hence G is not cyclic. $\square$

Example 3.1: $\mathbb{Z}_{20} \cong \mathbb{Z}_4 \times \mathbb{Z}_5$ (from lecture)

By Theorem 3.1: $\gcd(4, 5) = 1$, so $\mathbb{Z}_4 \times \mathbb{Z}_5$ is cyclic of order 20, hence $\mathbb{Z}_4 \times \mathbb{Z}_5 \cong \mathbb{Z}_{20}$.

But $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5$: $\gcd(2, 2) = 2 \neq 1$, so it is not cyclic, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \ncong \mathbb{Z}_{20}$.

4 Classification Theorem

Theorem 4.1: Classification of finite abelian groups (from lecture; cf. Lee, Thm. 5.6)

Let G and H be finite abelian groups. Then $G \cong H$ if and only if G and H have the same elementary divisors.

Proof. See Lee, Theorem 5.6. □

Remark 4.1: Consequence (from lecture)

To determine whether two finite abelian groups are isomorphic, compute their elementary divisors. If the lists agree (as multisets), the groups are isomorphic; otherwise they are not.

Example 4.1: $G \cong K \ncong H$ (from lecture; cf. Lee, Example 5.13)

Compute the elementary divisors of three abelian groups of order $9600 = 2^7 \times 3 \times 5^2$:

$G = \mathbb{Z}_{200} \times \mathbb{Z}_8 \times \mathbb{Z}_6$.

- $200 = 2^3 \times 5^2$: $\mathbb{Z}_{200} \cong \mathbb{Z}_8 \times \mathbb{Z}_{25}$
- $8 = 2^3$: stays $\mathbb{Z}_8$
- $6 = 2 \times 3$: $\mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$

So $G \cong \mathbb{Z}_8 \times \mathbb{Z}_{25} \times \mathbb{Z}_8 \times \mathbb{Z}_2 \times \mathbb{Z}_3$.

Elementary divisors: $\{8, 8, 2, 3, 25\}$.

$H = \mathbb{Z}_{120} \times \mathbb{Z}_{10} \times \mathbb{Z}_4 \times \mathbb{Z}_2$.

- $120 = 2^3 \times 3 \times 5$: $\mathbb{Z}_{120} \cong \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_5$
- $10 = 2 \times 5$: $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$
- $4 = 2^2$: stays $\mathbb{Z}_4$
- 2: stays $\mathbb{Z}_2$

So $H \cong \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_5 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_4 \times \mathbb{Z}_2$.

Elementary divisors: $\{8, 4, 2, 2, 3, 5, 5\}$.

$K = \mathbb{Z}_{25} \times \mathbb{Z}_{24} \times \mathbb{Z}_8 \times \mathbb{Z}_2$.

- $25 = 5^2$: stays $\mathbb{Z}_{25}$
- $24 = 2^3 \times 3$: $\mathbb{Z}_{24} \cong \mathbb{Z}_8 \times \mathbb{Z}_3$
- $8 = 2^3$: stays $\mathbb{Z}_8$
- 2: stays $\mathbb{Z}_2$

So $K \cong \mathbb{Z}_{25} \times \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_8 \times \mathbb{Z}_2$.

Elementary divisors: $\{25, 8, 3, 8, 2\} = \{8, 8, 2, 3, 25\}$.

Conclusion. G and K have the same elementary divisors $\{8, 8, 2, 3, 25\}$, so $G \cong K$. H has elementary divisors $\{8, 4, 2, 2, 3, 5, 5\} \neq \{8, 8, 2, 3, 25\}$, so $H \ncong G$ and $H \ncong K$.

Checklist of Skills

- ☐ Prove the FTFAG: cite Lemma 6 to reduce to p-groups, then use strong induction with Lemma 7 to peel off cyclic factors.
- ☐ Define elementary divisors and state why they are invariants of G.
- ☐ List all abelian groups of a given order by writing all ways to factor the order as products of prime powers, matching the structure of partitions.
- ☐ Explain why $\mathbb{Z}_{20}$ does not appear separately for order 20 (it equals $\mathbb{Z}_4 \times \mathbb{Z}_5$ via the cyclic product criterion).
- ☐ State and prove Theorem 3.1 (cyclic product criterion) both directions.
- ☐ In the ($\Leftarrow$) direction: use coprimality to show the product element has order $|G|$.
- ☐ In the ($\Rightarrow$) direction: use the prime dividing two n_i's to bound every element order below $|G|$.
- ☐ State the Classification Theorem (Lee Thm. 5.6): $G \cong H$ iff same elementary divisors.
- ☐ Compute elementary divisors of a given group (factor each cyclic factor into prime power components, collect the prime powers).
- ☐ Use elementary divisors to determine isomorphism between groups.